

**IDENTIFICATION**CODE : GI-4-S1-EC-OEA  
ECTS : 2**HOURS**Cours : 0h  
TD : 0h  
TP : 28h  
Projet : 0h  
Evaluation : 0h  
Face à face pédagogique : 28h  
Travail personnel : 12h  
Total : 40h**ASSESSMENT METHOD**

Lab: 100 %

**TEACHING AIDS**

Slides et Polycopié

**TEACHING LANGUAGE**

French

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This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

Provide the basis of operational research (concepts, methods, models) for industrial engineers.

Enable the students to master linear programming, integer linear programming, graph theory, to model and solve industrial and logistic problems, using optimization softwares.

**CONTENT**

The first half of the course alternates between 2-hour lectures and practical sessions, introducing the main concepts of exact optimization and key metaheuristics. It covers a review of linear programming and the simplex method, as well as complexity, integer linear programming, path-based heuristics, and population-based heuristics.

The second half of the course is dedicated to a project. In 2024-25, it focused on a two-tier urban logistics problem, where a truck tour supplies depots, and bicycles then collect goods from the depots to deliver them to customers' homes. A code implementing an algorithmic solution to the problem is provided. Students are then required to replace the solution components with more advanced exact or heuristic methods.

**BIBLIOGRAPHY****PRE-REQUISITES**Probability and statistics  
Algorithmics  
Programming  
Integer Linear programming